

Text Processing Assessment Report

Username: LIR23GK

This report compares the **Naive-bayes** and **Dictionary-based** classifiers for sentiment analysis using accuracy, precision, recall, and F1 metrics. It also evaluates enhancements to the dictionary-based model, performs error analysis, and identifies avenues for improvement in cross-domain adaptability.

Methodology and Results

Step 2: Training and Testing Naive-bayes Classifier

The Naive-bayes classifier was trained on the Rotten Tomatoes dataset and evaluated on the test set to assess its performance in sentiment classification.

2.1 Processing Sentiment Dictionaries

- Modified the code to store the processed words in lists *posWordList* and *negWordList* by removing the leading and trailing white spaces, skipping the lines that are blank or start with a semicolon(;) which are comments in the dictionary files.

2.2 Elevation Metrics Function

- A custom common function ***calculate_metrics*** is implemented to calculate the following metrics:
 - Accuracy: Proportion of correctly predicted label
 - Precision: Proportion of positive predictions that are correct
 - Recall: Proportion of actual positives that are correctly predicted
 - F1-Measure: Harmonic mean of precision and recall

2.3 Performance Metrics

The classifier performance on the test set was evaluated, and the following metrics were computed:

Metrics	Positive Sentiment	Negative Sentiment
F-Measure	0.7705	0.7792
Precision	0.7882	0.7627
Recall	0.7536	0.7963
Accuracy	0.7749	

The model favors negative sentiment recall, with positive sentiment favoring precision. Misclassification occurs with ambiguity, sarcasm, or context-dependent phrases.

Step 3: Testing Naive-bayes on Nokia Dataset

3.1 Metrics Captured for Films and Nokia reviews Naive-bayes

Film Reviews (Train Data)

Metrics	Positive Sentiment	Negative Sentiment
F-Measure	0.8904	0.8919
Precision	0.8998	0.8828
Recall	0.8811	0.9012
Accuracy	0.8911	

Film Reviews (*Test Data*)

Metrics	Positive Sentiment	Negative Sentiment
F-Measure	0.7739	0.7842
Precision	0.7671	0.7909
Recall	0.7809	0.7776
Accuracy	0.7792	

Nokia Reviews (*All Data*)

Metrics	Positive Sentiment	Negative Sentiment
F-Measure	0.6708	0.4952
Precision	0.7941	0.4000
Recall	0.5806	0.6500
Accuracy	0.6015	

The model's accuracy dropped significantly on Nokia reviews (60.15%), reflecting a lack of adaptability to new domains.

3.2 Naive-bayes: Cross-Domain Evaluation

The Naive-bayes classifier performed well on the film review training data with an accuracy of 89.11% and consistently high F-measure, precision, and recall for both positive and negative sentiments. This is an indication that the model has been able to learn sentiment patterns from the training data quite well. However, the accuracy fell to 77.92% on unseen film reviews, while there was also a slight drop in F-measure and other metrics. It also reflects the model's general weakness in generalising well out of the training data set-a well-entrenched problem for all machine learning models after exposing them to new examples.

On the Nokia product reviews, the classifier resulted in an even poorer accuracy, that is, just 60.15%. It can be realised that it has the high probability of facing a low value

of accuracy when this sentiment model was applied across other domains. The F-measure was very low for negative sentiment, 0.4952, with poor precision, 0.40, indicating high misclassification. These results hint that the classifier suffers from domain adaptation, and therefore models should be tuned to a specific dataset or techniques considering cross-domain variability should be used.

Step 4: Identifying and Evaluating Most Useful Words

4.1 Appendix I : Most Useful Words for Predicting Sentiment

Above Appendix refers to the words with highest predictive value for both positive and negative sentiment.

4.2 Appendix II: Overlap Words

The model identified 91 words (49 positive, 42 negative) from the sentiment dictionary as most useful, indicating a strong reliance on predefined sentiment terms. This shows that the model effectively utilises traditional sentiment-bearing words for classification. However, the presence of additional domain-specific terms among the most useful words suggests that the model also learns context-specific vocabulary, enhancing its predictive power. The overlap highlights the strength of the sentiment dictionary while demonstrating the model's adaptability to domain-specific contexts, achieving a balance between traditional and contextual terms. This balance enhances the Naive-bayes model's robustness in sentiment classification tasks.

Step 5: Enhancing the Dictionary-based Classifier

5.1 Metrics of testDictionary() Dictionary-based

Films (Train Data)

Metrics	Positive Sentiment	Negative Sentiment
F-Measure	0.6221	0.6797
Precision	0.6828	0.6320
Recall	0.5713	0.7351
Accuracy	0.6533	

Films (Test Data)

Metrics	Positive Sentiment	Negative Sentiment
F-Measure	0.6144	0.6708
Precision	0.6773	0.6215
Recall	0.5623	0.7285
Accuracy	0.6448	

Nokia (*All Data*)

Metrics	Positive Sentiment	Negative Sentiment
F-Measure	0.8492	0.6897
Precision	0.8837	0.6383
Recall	0.8172	0.7500
Accuracy	0.7970	

5.2 Performance of Enhanced Dictionary-based Classifier

Films (*Train and Test Data*):

The dictionary-based approach performs reasonably well but is generally less accurate than the Naive-bayes classifier on both training and test data.

- **Accuracy:**
 - Films (Train Data): **65.33%** (Dictionary) vs. **89.11%** (Naive-bayes).
 - Films (Test Data): **64.48%** (Dictionary) vs. **77.92%** (Naive-bayes).
- **Precision and Recall:**
 - The dictionary-based approach struggles with recall for positive sentiment, suggesting it misses many positive examples. Naive-bayes achieve better balance across metrics.

Nokia Product Reviews:

Surprisingly, the dictionary-based approach achieves a higher accuracy (**79.70%**) on the Nokia dataset compared to its performance on film reviews. This may indicate that the sentiment dictionary aligns better with product review terms.

However, Naive-bayes, when trained on film reviews, performs poorly on Nokia reviews due to the domain shift. The dictionary-based method shows better adaptability to new domains but still lacks the statistical nuance of Naive-bayes.

5.3 Metrics of enhancedTestDictionary() *Dictionary-based*

This section evaluates the performance improvements achieved through the enhancements made to the dictionary-based classifier, focusing on negation handling and diminisher rules.

Films (*Train Data*)

Metrics	Positive Sentiment	Negative Sentiment
F-Measure	0.6146	0.6792
Precision	0.6875	0.6241

Recall	0.5556	0.7450
Accuracy	0.6499	

Films (*Test Data*)

Metrics	Positive Sentiment	Negative Sentiment
F-Measure	0.6147	0.6833
Precision	0.6514	0.6531
Recall	0.5820	0.7164
Accuracy	0.6523	

Nokia (*All Data*)

Metrics	Positive Sentiment	Negative Sentiment
F-Measure	0.8497	0.7204
Precision	0.9187	0.6321
Recall	0.7903	0.8375
Accuracy	0.8108	

The enhanced dictionary-based function (***enhancedTestDictionary()***) demonstrated improved performance, especially in the Nokia dataset, compared to the basic dictionary-based approach. Below are the key observations based on the metrics:

Films Dataset (Train and Test Data):

Train Data:

- The accuracy was slightly lower, 64.99%, compared to the basic dictionary-based function, 65.33%, showing that improvements did not do much in the reduction of misclassifications.
- Positive sentiment precision increased slightly, 0.6875 versus 0.6828, but recall remained low at 0.5556, indicating difficulties in detecting subtle positive expressions.
- Negative sentiment recall (0.7450 vs. 0.7351) saw slight improvement, likely due to better negation handling, though precision gains were minimal.

Test Data:

- It achieved an accuracy of 65.23% compared to the basic model's 64.48%, reflecting minor improvements in generalization.

- Positive sentiment precision at 0.6514 increased, reflecting fewer false positives for this class; the recall of negative sentiments at 0.7164 improved, reflecting a better way of handling ambiguous or context-sensitive phrases.

Observations:

- Negation handling and diminisher rules had limited impact due to low domain complexity, highlighting the inflexibility of dictionary-based approaches compared to Naive-bayes for domain-agnostic tasks like film reviews.

Nokia Dataset:

Performance Improvements:

- Accuracy reached 81.08%, a notable improvement over both the basic dictionary-based approach (79.70%) and Naive-bayes (60.15%), demonstrating the importance of targeted enhancements in domain-specific datasets.
- Positive sentiment precision (0.9187 vs. 0.8837) saw significant improvement, reflecting better classification of strongly positive phrases. However, recall (0.7903 vs. 0.8172) reduced slightly, showing a trade-off in identifying less obvious positive instances.
- Negative sentiment recall increased markedly to 0.8375 (vs. 0.7500), attributed to better handling of negations (e.g., “not good”) and context-sensitive terms. Precision remained stable at 0.7204.

Observations:

- The enhanced function outperforms Naive-bayes on Nokia data, leveraging domain-specific vocabulary, negation, and diminisher rules, proving rule-based approaches excel with clear, domain-aware sentiment markers.

Step 6: Error Analysis

6.3 Error Patterns

[Appendix III: Naive-bayes Classifications on Film Reviews Test Data](#)

[Appendix IV: Test Dictionary Classification on Nokia Reviews Dataset](#)

6.4 Limitations and Future Improvements

The **Naive-bayes model** demonstrates strong domain-specific learning but struggles with sentences that are context-dependent, sarcastic, or contain mixed sentiments. Positive sentences misclassified as negative often include subtle positivity or negation that the model cannot interpret effectively, such as “not a bad choice here.” Similarly, negative sentences misclassified as positive frequently contain positive keywords in an overall negative context, as seen in “a humorless journey into a philosophical void.” These errors highlight the model’s reliance on individual word probabilities and its inability to grasp the broader context or tone of the sentences.

The **Dictionary-based model**, on the other hand, is more interpretable but overly dependent on predefined keywords, leading to frequent misclassifications when explicit sentiment is absent. Positive sentences without strong sentiment words, such as “it is small,” are often classified as negative. Conversely, negative sentences with positive words, like “great battery life, but a tid bit quieter than I would like,” result in errors due to a lack of contextual understanding. This approach is limited in handling

complex linguistic structures like negation, mixed sentiments, or domain-specific critiques, making it less adaptable to nuanced or ambiguous language.

Appendix I: Most Useful Words for Predicting Sentiment

NEGATIVE:

['generic', 'mediocre', 'badly', 'routine', 'ill', 'unfunny', 'poorly', 'mindless', 'boring', 'stale', 'pointless', 'bore', 'dreary', 'annoying', 'shoot', 'disguise', 'meandering', "wasn't", 'nowhere', 'tiresome', 'disaster', 'stupid', 'unless', 'junk', 'waste', 'apparently', 'banal', 'fatal', 'offensive', 'plotting', 'product', 'seagal', 'lousy', 'harvard', 'designed', 'pinocchio', 'flat', 'dull', 'numbers', 'animal', 'amateurish', 'horrible', 'pathetic', 'sadly', 'ballistic', 'plodding', 'lifeless', 'conceived', 'supposed', 'stealing', 'preachy', 'cable', 'chan', 'sour', 'kung', 'trite', 'wasted', 'pokemon', 'unintentional', 'incoherent', 'stiff', 'choppy', 'uninspired', 'stunt', 'soggy', 'suffers', 'lame', 'unintentionally', 'hollow', 'inept', 'store', 'unfocused', 'intentioned', 'obnoxious', 'meant', 'witless', 'santa', 'missed', 'putting', 'total', 'crude', 'clumsily', 'leaden', 'bother', 'sara', 'inane', 'benigni', '51', 'crap', 'ship', 'sinks', 'wannabe', '8', 'saccharine', 'serving', 'imitation', 'settles', 'contains', 'arts', 'relentlessly']

POSITIVE:

['ingenious', 'jealousy', 'spite', 'gradually', 'superbly', 'gently', 'para', 'filme', 'deadpan', 'harrowing', 'desperation', 'delightful', 'rare', 'vivid', 'grandeur', 'masterful', 'bringing', 'buoyant', 'lovers', 'wrenching', 'format', 'potent', 'uncompromising', 'droll', 'heartbreaking', 'timely', 'absorbing', 'intimate', 'speaks', 'affecting', 'pianist', 'sly', 'integrity', 'washington', 'nuanced', 'delightfully', 'richly', 'subversive', 'skin', 'intoxicating', 'joyous', 'unique', 'lane', 'undeniably', 'bourne', 'russian', 'honesty', 'unexpected', 'polished', 'transcends', 'frailty', 'explores', 'record', 'hopeful', 'smarter', 'sadness', 'warm', 'evocative', 'martha', 'son', 'iranian', 'resonant', 'captivating', 'sides', 'assured', 'consistently', 'powerful', 'captures', 'flaws', 'pulls', 'detailed', 'aspects', 'grown', 'touching', 'document', 'wry', 'provides', 'tender', 'playful', 'lively', 'tour', 'spare', 'respect', 'vividly', 'realistic', 'heartwarming', 'wonderfully', 'wonderful', 'refreshingly', 'thoughtful', 'riveting', 'mesmerizing', 'ages', 'chilling', 'gem', 'refreshing', 'haunting', 'inventive', 'ya', 'engrossing']

Appendix II: Overlap Words

Number of most useful negative words in sentiment dictionary: 42

Number of most useful positive words in sentiment dictionary: 49

Total count of most useful words in sentiment dictionary: 91

Appendix III: Naive-bayes Classifications on Film Reviews Test Data

testBayes ERROR (pos classed as neg 0.45):a screenplay more ingeniously constructed than " memento "

testBayes ERROR (pos classed as neg 0.01):4 friends , 2 couples , 2000 miles , and all the pabst blue ribbon beer they can drink - it's the ultimate redneck road-trip .

testBayes ERROR (pos classed as neg 0.08):though this film can be clumsy , its ambitions are equally -- and admirably -- uncommercial .

testBayes ERROR (pos classed as neg 0.16):day is not a great bond movie , but it is a good bond movie , which still makes it much better than your typical bond knock-offs .

testBayes ERROR (pos classed as neg 0.08):legendary irish writer brendan behan's memoir , borstal boy , has been given a loving screen transferral .

testBayes ERROR (pos classed as neg 0.22):spielberg is the rare director who does not want to invite viewers to gawk at or applaud his special effects . he just wants them to be part of the action , the wallpaper of his chosen reality . here , thankfully , they are .

testBayes ERROR (pos classed as neg 0.39):an involving true story of a chinese actor who takes up drugs and winds up in an institution--acted mostly by the actual people involved .

testBayes ERROR (pos classed as neg 0.03):while not all transitions to adulthood are so fraught , there's much truth and no small amount of poetry in girls can't swim .

testBayes ERROR (pos classed as neg 0.00):despite the long running time , the pace never feels slack -- there's no scene that screams " bathroom break ! "

testBayes ERROR (pos classed as neg 0.00):the whole cast looks to be having so much fun with the slapstick antics and silly street patois , tossing around obscure expressions like bellini and mullinski , that the compact 86 minutes breezes by .

testBayes ERROR (pos classed as neg 0.35):notorious c . h . o . has oodles of vulgar highlights .

testBayes ERROR (pos classed as neg 0.36):what it lacks in originality it makes up for in intelligence and b-grade stylishness .

testBayes ERROR (pos classed as neg 0.01):bow's best moments are when he's getting busy on the basketball court because that's when he really scores .

testBayes ERROR (pos classed as neg 0.18):don't expect any surprises in this checklist of teamwork cliches . . .

testBayes ERROR (pos classed as neg 0.13):just another fish-out-of-water story that barely stays afloat .

testBayes ERROR (pos classed as neg 0.06):it's hard not to be seduced by [witherspoon's] charisma , even in this run-of-the-mill vehicle , because this girl knows how to drive it to the max .

testBayes ERROR (pos classed as neg 0.21):a miniscule little bleep on the film radar , but one that many more people should check out

testBayes ERROR (pos classed as neg 0.39):full frontal is the antidote for soderbergh fans who think he's gone too commercial since his two oscar nominated films in 2000

testBayes ERROR (pos classed as neg 0.03):[hawn's character]is so bluntly written , without a trace of sentimentality , and so blisteringly defined , that every other character seems overlooked and underwritten .

testBayes ERROR (pos classed as neg 0.04):this isn't a narrative film -- i don't know if it's possible to make a narrative film about september 11th , though i'm sure some will try -- but it's as close as anyone has dared to come .

testBayes ERROR (pos classed as neg 0.23):never once predictable .

testBayes ERROR (pos classed as neg 0.35):if the count of monte cristo doesn't transform caviezel into a movie star , then the game is even more rigged than it was two centuries ago .

testBayes ERROR (pos classed as neg 0.11):one of creepiest , scariest movies to come along in a long , long time , easily rivaling blair witch or the others .

testBayes ERROR (pos classed as neg 0.01):it seems like i have been waiting my whole life for this movie and now i can't wait for the sequel .

testBayes ERROR (pos classed as neg 0.07):if you've ever wondered what an ending without the input of studio executives or test audiences would look like , here it is .

testBayes ERROR (pos classed as neg 0.00):opening with some contrived banter , cliches and some loose ends , the screenplay only comes into its own in the second half .

testBayes ERROR (pos classed as neg 0.00):shamelessly resorting to pee-related sight gags that might even cause tom green a grimace ; still , myer's energy and the silliness of it all eventually prevail

testBayes ERROR (pos classed as neg 0.00):too damn weird to pass up , and for the blacklight crowd , way cheaper (and better) than pink floyd tickets .

testBayes ERROR (pos classed as neg 0.31):you'd have to be a most hard-hearted person not to be moved by this drama .

testBayes ERROR (pos classed as neg 0.15):there are no special effects , and no hollywood endings .

testBayes ERROR (pos classed as neg 0.01):despite its hawaiian setting , the science-fiction trimmings and some moments of rowdy slapstick , the basic plot of " lilo " could have been pulled from a tear-stained vintage shirley temple script .

testBayes ERROR (pos classed as neg 0.00):you might want to take a reality check before you pay the full ticket price to see " simone , " and consider a dvd rental instead .

testBayes ERROR (pos classed as neg 0.03):there's no reason to miss interview with the assassin

testBayes ERROR (pos classed as neg 0.31):the movie is well done , but slow .

testBayes ERROR (pos classed as neg 0.08):the filmmakers want nothing else than to show us a good time , and in their cheap , b movie way , they succeed .

testBayes ERROR (pos classed as neg 0.08):a positively thrilling combination of ethnography and all the intrigue , betrayal , deceit and murder of a shakespearian tragedy or a juicy soap opera .

testBayes ERROR (pos classed as neg 0.37):just about the best straight-up , old-school horror film of the last 15 years .

testBayes ERROR (pos classed as neg 0.19):if you want to see a train wreck that you can't look away from , then look no further , because here it is .

testBayes ERROR (pos classed as neg 0.18):combine the paranoid claustrophobia of a submarine movie with the unsettling spookiness of the supernatural -- why didn't hollywood think of this sooner ?

testBayes ERROR (pos classed as neg 0.03):the plot twists give i am trying to break your heart an attraction it desperately needed .

testBayes ERROR (pos classed as neg 0.16):muccino seems to be exploring the idea of why human beings long for what they don't have , and how this gets us in trouble . but even while his characters are acting horribly , he is always sympathetic .

testBayes ERROR (pos classed as neg 0.47):the results , if not memorable , are at least interesting .

testBayes ERROR (pos classed as neg 0.01):while parker and co-writer catherine di napoli are faithful to melville's plotline , they and a fully engaged supporting cast . . . have made the old boy's characters more quick-witted than any english lit major would have thought possible .

testBayes ERROR (pos classed as neg 0.05):sometimes this 'blood' seems as tired as its protagonist . . . still , the pulse never disappears entirely , and the picture crosses the finish line winded but still game .

testBayes ERROR (pos classed as neg 0.15):this is more fascinating -- being real -- than anything seen on jerry springer .

testBayes ERROR (pos classed as neg 0.03):a different movie -- sometimes tedious -- by a director many viewers would like to skip but film buffs should get to know .

testBayes ERROR (pos classed as neg 0.21):a much better documentary -- more revealing , more emotional and more surprising -- than its pedestrian english title would have you believe .

testBayes ERROR (pos classed as neg 0.29):'like a child with an important message to tell . . . [skins'] faults are easy to forgive because the intentions are lofty . '

testBayes ERROR (pos classed as neg 0.02):the ring is worth a look , if you don't demand much more than a few cheap thrills from your halloween entertainment .

testBayes ERROR (pos classed as neg 0.12):at times a bit melodramatic and even a little dated (depending upon where you live) , ignorant fairies is still quite good-natured and not a bad way to spend an hour or two .

testBayes ERROR (pos classed as neg 0.13):maybe it is formula filmmaking , but there's nothing wrong with that if the film is well-crafted and this one is .

testBayes ERROR (pos classed as neg 0.33):lavishly , exhilaratingly tasteless .

testBayes ERROR (pos classed as neg 0.46):gooding is the energetic frontman , and it's hard to resist his enthusiasm , even if the filmmakers come up with nothing original in the way of slapstick sequences .

testBayes ERROR (pos classed as neg 0.43):the stars may be college kids , but the subject matter is as adult as you can get : the temptations of the flesh are unleashed by a slightly crazed , overtly determined young woman and a one-night swim turns into an ocean of trouble .

testBayes ERROR (pos classed as neg 0.31):talk to her is so darned assured , we have absolutely no idea who the main characters are until the film is well under way -- and yet it's hard to stop watching .

testBayes ERROR (pos classed as neg 0.32):while guzmán frustratingly refuses to give pinochet's crimes a political context , his distance from the material is mostly admirable .

testBayes ERROR (pos classed as neg 0.10):not a bad choice here , assuming that . . . the air-conditioning in the theater is working properly .

testBayes ERROR (pos classed as neg 0.10):the subtitled costume drama is set in a remote african empire before cell phones , guns , and the internal combustion engine , but the politics that thump through it are as timely as tomorrow .

testBayes ERROR (pos classed as neg 0.39):director douglas mcgrath takes on nickleby with all the halfhearted zeal of an 8th grade boy delving into required reading .

testBayes ERROR (pos classed as neg 0.03):elvira fans could hardly ask for more .

testBayes ERROR (pos classed as neg 0.16):not the best herzog perhaps , but unmistakably herzog .

testBayes ERROR (pos classed as neg 0.40):the one-liners are snappy , the situations volatile and the comic opportunities richly rewarded .

testBayes ERROR (pos classed as neg 0.06):you may think you have figured out the con and the players in this debut film by argentine director fabian bielinsky , but while you were thinking someone made off with your wallet .

testBayes ERROR (pos classed as neg 0.32):a fantastic premise anchors this movie , but what it needs is either a more rigid , blair witch-style commitment to its mockumentary format , or a more straightforward , dramatic treatment , with all the grandiosity that that implies .

testBayes ERROR (pos classed as neg 0.30):hailed as a clever exercise in neo-hitchcockianism , this clever and very satisfying picture is more accurately chabrolian .

testBayes ERROR (pos classed as neg 0.10):be prepared to cling to the edge of your seat , tense with suspense . the ring never lets you off the hook .

testBayes ERROR (pos classed as neg 0.04):even these tales of just seven children seem at times too many , although in reality they are not enough . every child's story is what matters . this film can only point the way -- but thank goodness for this signpost .

testBayes ERROR (pos classed as neg 0.43):although some viewers will not be able to stomach so much tongue-in-cheek weirdness , those who do will have found a cult favorite to enjoy for a lifetime .

testBayes ERROR (pos classed as neg 0.09):if you like blood , guts and crazy beasts stalking men with guns though . . . you will likely enjoy this monster .

testBayes ERROR (pos classed as neg 0.20):' . . . the film's considered approach to its subject matter is too calm and thoughtful for agitprop , and the thinness of its characterizations makes it a failure as straight drama . '

testBayes ERROR (pos classed as neg 0.06):the screenwriters dig themselves in deeper every time they toss logic and science into what is essentially a " dungeons and dragons " fantasy with modern military weaponry . . .

testBayes ERROR (pos classed as neg 0.49):good-naturedly cornball sequel .

testBayes ERROR (pos classed as neg 0.37):it's quite diverting nonsense .

testBayes ERROR (pos classed as neg 0.36):the directive to protect the code at all costs also begins to blur as the importance of the man and the code merge

testBayes ERROR (pos classed as neg 0.35):feeling like a dope has rarely been more fun than it is in nine queens .

testBayes ERROR (pos classed as neg 0.15):what saves it . . . and makes it one of the better video-game-based flicks , is that the film acknowledges upfront that the plot makes no sense , such that the lack of linearity is the point of emotional and moral departure for protagonist alice .

testBayes ERROR (pos classed as neg 0.23):a naturally funny film , home movie makes you crave chris smith's next movie .

testBayes ERROR (pos classed as neg 0.47):sheds light on a subject few are familiar with , and makes you care about music you may not have heard before .

testBayes ERROR (pos classed as neg 0.46):sure , it's contrived and predictable , but its performances are so well tuned that the film comes off winningly , even though it's never as solid as you want it to be .

testBayes ERROR (pos classed as neg 0.06):the santa clause 2 proves itself a more streamlined and thought out encounter than the original could ever have hoped to be .

testBayes ERROR (pos classed as neg 0.49):'enigma' is the kind of engaging historical drama that hollywood appears to have given up on in favor of sentimental war movies in the vein of 'we were soldiers . '

testBayes ERROR (pos classed as neg 0.46):sparkling , often hilarious romantic jealousy comedy . . . attal looks so much like a young robert deniro that it seems the film should instead be called 'my husband is travis bickle' .

testBayes ERROR (pos classed as neg 0.19):what a concept , what an idea , what a thrill ride . this is a more fascinating look at the future than " bladerunner " and one of the most high-concept sci fi adventures attempted for the screen .

testBayes ERROR (pos classed as neg 0.01):many went to see the attraction for the sole reason that it was hot outside and there was air conditioning inside , and i don't think that a . c . will help this movie one bit .

testBayes ERROR (pos classed as neg 0.10):disappointingly , the characters are too strange and dysfunctional , tom included , to ever get under the skin , but this is compensated in large part by the off-the-wall dialogue , visual playfulness and the outlandishness of the idea itself .

testBayes ERROR (pos classed as neg 0.40):a mildly enjoyable if toothless adaptation of a much better book .

testBayes ERROR (pos classed as neg 0.38):while the plot follows a predictable connect-the-dots course . . . director john schultz colors the picture in some evocative shades .

testBayes ERROR (pos classed as neg 0.39): . . . a somber film , almost completely unrelieved by any comedy beyond the wistful everyday ironies of the working poor .

testBayes ERROR (pos classed as neg 0.14):while cherish doesn't completely survive its tonal transformation from dark comedy to suspense thriller , it's got just enough charm and appealing character quirks to forgive that still serious problem .

testBayes ERROR (pos classed as neg 0.42):the entire cast is first-rate , especially sorvino .

testBayes ERROR (pos classed as neg 0.04):like the best 60 minutes exposé , the film (at 80 minutes) is actually quite entertaining .

testBayes ERROR (pos classed as neg 0.03):claire is a terrific role for someone like judd , who really ought to be playing villains .

testBayes ERROR (pos classed as neg 0.46):it will break your heart many times over .

testBayes ERROR (pos classed as neg 0.42):the dirty jokes provide the funniest moments in this oddly sweet comedy about jokester highway patrolmen .

testBayes ERROR (pos classed as neg 0.00):even before it builds up to its insanely staged ballroom scene , in which 3000 actors appear in full regalia , it's waltzed itself into the art film pantheon .

testBayes ERROR (pos classed as neg 0.38):the salton sea has moments of inspired humour , though every scrap is of the darkest variety .

testBayes ERROR (pos classed as neg 0.08):a very pretty after-school special . it's an effort to watch this movie , but it eventually pays off and is effective if you stick with it .

testBayes ERROR (pos classed as neg 0.00):neither the funniest film that eddie murphy nor robert de niro has ever made , showtime is nevertheless efficiently amusing for a good while . before it collapses into exactly the kind of buddy cop comedy it set out to lampoon , anyway .

testBayes ERROR (pos classed as neg 0.03):you can fire a torpedo through some of clancy's holes , and the scripters don't deserve any oscars . but the nerve-raked acting , the crackle of lines , the impressive stagings of hardware , make for some robust and scary entertainment .

testBayes ERROR (pos classed as neg 0.08):you can take the grandkids or the grandparents and never worry about anyone being bored . . . audience is a sea of constant smiles and frequent laughter .

testBayes ERROR (pos classed as neg 0.15):though the film never veers from its comic course , its unintentional parallels might inadvertently evoke memories and emotions which are anything but humorous .

testBayes ERROR (pos classed as neg 0.31):one of those movies that catches you up in something bigger than yourself , namely , an archetypal desire to enjoy good trash every now and then .

testBayes ERROR (pos classed as neg 0.11):apart from anything else , this is one of the best-sustained ideas i have ever seen on the screen .

testBayes ERROR (pos classed as neg 0.06):were it not for a sentimental resolution that explains way more about cal than does the movie or the character any good , freundlich's world traveler might have been one of the more daring and surprising american movies of the year .

testBayes ERROR (pos classed as neg 0.33):we've seen the hippie-turned-yuppie plot before , but there's an enthusiastic charm in <i ? fire that makes the formula fresh again .

testBayes ERROR (pos classed as neg 0.49):even if the naipaul original remains the real masterpiece , the movie possesses its own languorous charm .

testBayes ERROR (pos classed as neg 0.48):by its modest , straight-ahead standards , undisputed scores a direct hit .

testBayes ERROR (pos classed as neg 0.25):while the story does seem pretty unbelievable at times , it's awfully entertaining to watch .

testBayes ERROR (pos classed as neg 0.10):it's like rocky and bullwinkle on speed , but that's neither completely enlightening , nor does it catch the intensity of the movie's strangeness .

testBayes ERROR (pos classed as neg 0.03):another in a long line of ultra-violent war movies , this one is not quite what it could have been as a film , but the story and theme make up for it .

testBayes ERROR (pos classed as neg 0.08):the modern remake of dumas's story is long on narrative and (too) short on action .

testBayes ERROR (pos classed as neg 0.18):the symbols float like butterflies and the spinning styx sting like bees . i wanted more .

testBayes ERROR (pos classed as neg 0.10):neither quite a comedy nor a romance , more of an impish divertissement of themes that interest attal and gainsbourg -- they live together -- the film has a lot of charm .

testBayes ERROR (pos classed as neg 0.46):the film may not hit as hard as some of the better drug-related pictures , but it still manages to get a few punches in .

testBayes ERROR (pos classed as neg 0.17):i'm happy to have seen it -- not as an alternate version , but as the ultimate exercise in viewing deleted scenes .

testBayes ERROR (pos classed as neg 0.01):the gags that fly at such a furiously funny pace that the only rip off that we were aware of was the one we felt when the movie ended so damned soon .

testBayes ERROR (neg classed as pos 0.61):beware the quirky brit-com . they can and will turn on a dime from oddly humorous to tediously sentimental .

testBayes ERROR (neg classed as pos 0.52):confusion is one of my least favourite emotions , especially when i have to put up with 146 minutes of it .

testBayes ERROR (neg classed as pos 0.78):fear permeates the whole of stortelling , todd solondz' oftentimes funny , yet ultimately cowardly autocritique .

testBayes ERROR (neg classed as pos 0.90): . . . ice age treads predictably along familiar territory , making it a passable family film that won't win many fans over the age of 12 .

testBayes ERROR (neg classed as pos 0.53):what you end up getting is the vertical limit of surfing movies - memorable stunts with lots of downtime in between .

testBayes ERROR (neg classed as pos 0.92):an inelegant combination of two unrelated shorts that falls far short of the director's previous work in terms of both thematic content and narrative strength .

testBayes ERROR (neg classed as pos 0.84):diane lane's sophisticated performance can't rescue adrian lyne's unfaithful from its sleazy moralizing .

testBayes ERROR (neg classed as pos 0.65):it's like an all-star salute to disney's cheesy commercialism .

testBayes ERROR (neg classed as pos 0.99):few of the increasingly far-fetched events that first-time writer-director neil burger follows up with are terribly convincing , which is a pity , considering barry's terrific performance .

testBayes ERROR (neg classed as pos 0.76):brady achieves the remarkable feat of squandering a topnotch foursome of actors . . . by shoving them into every clichéd white-trash situation imaginable .

testBayes ERROR (neg classed as pos 0.99):at no point during k-19 : the widowmaker did this viewer feel enveloped in a story that , though meant to be universal in its themes of loyalty , courage and dedication to a common goal , never seems to leave the lot .

testBayes ERROR (neg classed as pos 0.74): if the predictability of bland comfort food appeals to you , then the film is a pleasant enough dish .

testBayes ERROR (neg classed as pos 0.61):horrid little propaganda film with fascinating connections not only to the serbs themselves but also to a network of american right-wing extremists .

testBayes ERROR (neg classed as pos 0.82):consider the film a celluloid litmus test for the intellectual and emotional pedigree of your date and a giant step backward for a director i admire .

testBayes ERROR (neg classed as pos 0.90):chamber of secrets will find millions of eager fans . but if the essence of magic is its make-believe promise of life that soars above the material realm , this is the opposite of a truly magical movie .

testBayes ERROR (neg classed as pos 0.82):pure of intention and passably diverting , his secret life is light , innocuous and unremarkable .

testBayes ERROR (neg classed as pos 0.53):the film doesn't have enough innovation or pizazz to attract teenagers , and it lacks the novel charm that made spy kids a surprising winner with both adults and younger audiences .

testBayes ERROR (neg classed as pos 0.57):the very definition of what critics have come to term an " ambitious failure . "

testBayes ERROR (neg classed as pos 0.51):it's both sitcomishly predictable and cloying in its attempts to be poignant .

testBayes ERROR (neg classed as pos 0.57):uzumaki's interesting social parallel and defiant aesthetic seems a prostituted muse . . .

testBayes ERROR (neg classed as pos 0.95):'synthetic' is the best description of this well-meaning , beautifully produced film that sacrifices its promise for a high-powered star pedigree .

testBayes ERROR (neg classed as pos 0.62):it's impossible to indulge the fanciful daydreams of janice beard (eileen walsh) when her real-life persona is so charmless and vacant .

testBayes ERROR (neg classed as pos 0.74):self-congratulatory , misguided , and ill-informed , if nonetheless compulsively watchable .

testBayes ERROR (neg classed as pos 0.75):this sort of cute and cloying material is far from zhang's forte and it shows .

testBayes ERROR (neg classed as pos 0.98):even with harris's strong effort , the script gives him little to effectively probe ular's soul-stripping breakdown .

testBayes ERROR (neg classed as pos 0.67):most of the dialogue made me want to pack raw dough in my ears .

testBayes ERROR (neg classed as pos 0.57):may offend viewers not amused by the sick sense of humor .

testBayes ERROR (neg classed as pos 1.00):about amy's cuteness , amy's career success (she's a best-selling writer of self-help books who can't help herself) , and amy's neuroses when it comes to men .

testBayes ERROR (neg classed as pos 0.60):even murphy's expert comic timing and famed charisma can't rescue this effort .

testBayes ERROR (neg classed as pos 0.51):richard pryor mined his personal horrors and came up with a treasure chest of material , but lawrence gives us mostly fool's gold .

testBayes ERROR (neg classed as pos 0.97):plays like an unbalanced mixture of graphic combat footage and almost saccharine domestic interludes that are pure hollywood .

testBayes ERROR (neg classed as pos 0.99):the film doesn't show enough of the creative process or even of what was created for the non-fan to figure out what makes wilco a big deal .

testBayes ERROR (neg classed as pos 0.82):a sugar-coated rocky whose valuable messages are forgotten 10 minutes after the last trombone honks .

testBayes ERROR (neg classed as pos 0.80):new yorkers always seem to find the oddest places to dwell . . .

testBayes ERROR (neg classed as pos 0.59):welles groupie/scholar peter bogdanovich took a long time to do it , but he's finally provided his own broadside at publishing giant william randolph hearst .

testBayes ERROR (neg classed as pos 0.97):boasts eye-catching art direction but has a forcefully quirky tone that quickly wears out its limited welcome .

testBayes ERROR (neg classed as pos 0.97):yo , it's the days of our lives meets electric boogaloo .

testBayes ERROR (neg classed as pos 0.59):do we really need another film that praises female self-sacrifice ?

testBayes ERROR (neg classed as pos 0.54):a guilty pleasure at best , and not worth seeing unless you want to laugh at it .

testBayes ERROR (neg classed as pos 0.95):piccoli's performance is amazing , yes , but the symbols of loss and denial and life-at-arm's-length in the film seem irritatingly transparent .

testBayes ERROR (neg classed as pos 0.57):this is surely one of the most frantic , virulent and foul-natured christmas season pics ever delivered by a hollywood studio .

testBayes ERROR (neg classed as pos 0.78):don michael paul uses quick-cuts , (very) large shadows and wide-angle shots taken from a distance to hide the liberal use of a body double (for seagal) .

testBayes ERROR (neg classed as pos 0.60):the sight of the name bruce willis brings to mind images of a violent battlefield action picture , but the film has a lot more on its mind--maybe too much .

testBayes ERROR (neg classed as pos 0.55):once he starts learning to compromise with reality enough to become comparatively sane and healthy , the film becomes predictably conventional .

testBayes ERROR (neg classed as pos 1.00):[lee] treats his audience the same way that jim brown treats his women -- as dumb , credulous , unassuming , subordinate subjects , and lee seems just as expectant of an adoring , wide-smiling reception .

testBayes ERROR (neg classed as pos 0.87):swims in mediocrity , sticking its head up for a breath of fresh air now and then .

testBayes ERROR (neg classed as pos 0.50):proves a lovely trifle that , unfortunately , is a little too in love with its own cuteness .

testBayes ERROR (neg classed as pos 0.74):the campy results make mel brooks' borscht belt schtick look sophisticated .

testBayes ERROR (neg classed as pos 0.90):narc is all menace and atmosphere .

Appendix IV: Test Dictionary Classification on Nokia Reviews Dataset

testDictionary ERROR (pos classed as neg 0.00):i purchased this phone for just 50 \$ and there is no end to what i can do with it .

testDictionary ERROR (pos classed as neg 0.00):yes , the features are numerous .

testDictionary ERROR (pos classed as neg -1.00):i have not had any distortion problems with this phone and am more pleased with this phone than any i 've used before .

testDictionary ERROR (pos classed as neg 0.00):it is extremely light and dissapears in your pocket .

testDictionary ERROR (pos classed as neg 0.00):but anyway , the only thing you should really look at here is the fact that the phone does have voice dialing .

testDictionary ERROR (pos classed as neg 0.00):you can also assign special rings and graphics to special people when they call .

testDictionary ERROR (pos classed as neg 0.00):the phone also sports an image gallery (with option camera attachment) , a calendar / organizer , alarm clock with snooze , fm radio (way cool , even if it is just through an ear-peice) , games , java applications and gprs high-speed internet connection .

testDictionary ERROR (pos classed as neg 0.00):all my questions or problems have been quickly and satisfactorily resolved .

testDictionary ERROR (neg classed as pos 2.00):the colors on the screen are not as crisp as i 'd have liked them to be .

testDictionary ERROR (neg classed as pos 2.00):i have excellent hearing but the volume level on this phone is especially quiet .

testDictionary ERROR (neg classed as pos 2.00):since i received the phone , i spent countless hours on the phone with customer service reps who promised t-zones would work " in 24 hours " .

testDictionary ERROR (neg classed as pos 1.00):i must have heard this about a dozen times over the span of 2 weeks , when t-zones never worked .

testDictionary ERROR (neg classed as pos 1.00):the volume level of the phone is not all that good .

testDictionary ERROR (neg classed as pos 2.00):some of the higher pitched rings are very easy to hear , but not easy to listen to .

testDictionary ERROR (neg classed as pos 1.00):they are not wowable good .

testDictionary ERROR (neg classed as pos 1.00):so loud , really , that it does n't work terribly well as a silent ringer option .

testDictionary ERROR (neg classed as pos 1.00):- no option for caller-id pictures or individualized ringtones .

testDictionary ERROR (neg classed as pos 2.00):regarding pc software support nokia suite does not work with some versions of xp as i tried to connect my phone via infrared port on my laptop .

testDictionary ERROR (neg classed as pos 1.00):its quiet .

testDictionary ERROR (neg classed as pos 1.00):although i find it more convenient to use 1-touch dialing , this phone does not have voice dialing .

testDictionary ERROR (neg classed as pos 1.00):this model does have the traditional key arrangement , it 's just that they are really close to one another , and have unconventional shapes , so it takes a big getting used to for someone like me with big hands .

testDictionary ERROR (neg classed as pos 1.00):the internet functions of the phone - wap and gprs - will only work through t-mobile 's services , because they have deleted the menu options that would enable you to configure the phone to be used on a different network .

testDictionary ERROR (neg classed as pos 1.00):when talking the voice is not very clear .